

		ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		Assignment V	
Branch		ECE/EE/EEE	Programme	BTech	
Course Name		University Physics: Semiconductor Physics	Semester	III	
Course Code		PHY3009	Academic Year	2024-25	
Assignment-5		Topic: (BJT), MOSFET, etc			
Learning Level (LL)		L1: Remembering	L3: Applying	L5: Evaluating	
		L2: Understanding	L4: Analysing	L6: Creating	
Q's	Questions			COs	LL
1.	Explain the conditions of the forward active, cut off, and saturation modes in BJT			CO5	L2, L3
2.	Why is the base width made small compared to other two regions in BJT.			CO5	L2,L3
3.	Define the common base current gain and common emitter current gain. How are they related?			CO5	L2,L3
4.	Find out the common base current gain and common emitter current gain factors if collector current is 1mA and emitter current is 1.2 mA.			CO5	L2,L3
5.	What do you mean by base width modulation or Early effect?			CO5	L2, L3
6.	Sketch the energy band diagram in an MOS capacitor with an p- type substrate in accumulation, depletion and inversion modes.			CO6	L2, L3
7.	Find out the oxide capacitance (C_{ox}) of MOSFET if the thickness of silicon dioxide insulator is 300 Å			CO6	L3
8.	What is flat band voltage (V_{FB})? and threshold voltage (V_{TN})?			CO6	L2
9.	Consider silicon at $T = 300\text{ K}$ with fermi potential ($\phi_{fp}=0.347\text{V}$), calculate the maximum depletion width at threshold inversion point.			CO6	L3
10.	Describe the basic MOSFET structure and its operation with suitable diagram			CO6	L2
11.	Describe the working mechanism of photo Diode, and LED separately.			CO6	L2